

Radioindium and determination of neutron radial asymmetry for the PF-1000 plasma focus device

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Abstract Indium activation was used for radial neutron distribution measurements. The dense plasma focus device PF-1000 was used as the D–D neutron source. The geometry of the indium samples used to determine the neutron radial asymmetry (R_{as}) was beforehand optimized for sufficient γ -registration efficiency. The samples were placed on the vacuum vessel's outer wall, activated, and measured by γ -ray spectrometry. The experiment was supported by Monte Carlo calculations of the activation factor, allowing R_{as} to be calculated. The parameters R_{as} and the shift of neutron emission center introduced here are useful for describing D–D neutron emission distributions.

Keywords Activation technique · Radioindium · Plasma focus · Radial asymmetry

Introduction

Neutrons cannot be made. They have to be obtained from a source. Since Chadwick discovered neutron the rapid development of neutron sources has been observed as well as their application in various fields of human activity. A wide range of neutron sources are currently in use. The smallest and the cheapest sources are radioactive sources that use (α , n) reaction or sources that based on

spontaneous fission. The total neutron yield (Y_n) of this types of sources is also small. At the other end of this list one can easily find advanced technology machines like fission reactors and spallation sources. The neutron flux density for the above group easily reaches 10^{15} n cm⁻² s⁻¹ (ILL reactor, France) or even exceeds this value. For a special purpose highly radioactive isotopic source like ²⁵²Cf could be mounted. It costs $\sim 5 \times 10^5$ €.

The fast development of neutron sources based on light elements fusion has been observed during the last few decades. They are characterized by advanced, innovative technology, used for the construction of tokomaks, stellators, or even neutron generators. The D–D neutron emissivity of Join European Torus (JET) tokamak is in the range of 10^{17} neutrons per pulse meanwhile commercially available D–T neutron generators emit up to 10^{10} n s⁻¹ in a continuous or pulsed regime. The budget of international thermonuclear experimental reactor (ITER) tokamak (successor of JET) is planned for 10^{10} € while the cost of the commercially available neutron generator is in range of $\sim 2 \times 10^5$ € (depending on its configuration and equipment).

Off the beaten track to the mentioned group are plasma generators like Plasma Focus (PF). The PF-1000 device belongs to this category. This kind of neutron source can be easily made and the total cost of the current commercially available machine like PF-6 or PF-9 covers $\sim 2 \times 10^4$ €. They could emit 10^9 neutrons per pulse with a frequency of up to ~ 0.1 Hz. Also theirs exploitation costs are very limited. Thus the neutron from the PF machine is the cheapest one that can be obtained in mass production. It is worthy noting that PF machines are not suitable for energy production and neutron emission is not stable during series of pulses. But it is good enough for testing any neutron diagnostic for large plasma experiments.

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The PF-1000 device operated at the Institute of Plasma Physics and Laser Microfusion (IPPLM) is the largest PF experiment in the world. The hot, dense deuterium plasma that is formed inside the PF-1000's vacuum vessel is a powerful source of D–D neutrons. While some of these are thermonuclear, up to 95 % are the result of a beam–target interaction. Our recent studies of neutron angular distribution gave information about forward neutron anisotropy [1], which provides information regarding both the angular and energetic characteristics of the ion population responsible for the neutron production. Forward neutron anisotropy is a benchmark for the impact of the two biggest contributors to the total neutron emission yield [2]. Neutrons emitted in radial directions are considered to be thermonuclear in origin. These observations led to the conclusion that inferences of forward anisotropy and total Y_n based on the amount of neutrons measured in one radial direction can lead to incorrect results.

The radial asymmetry (R_{as}) of neutron emission is a source of information about the transverse distribution of neutron sources inside the plasma, combining their shape and the shift of the neutron emission center (SNEC) across the device axis. The activation technique is a powerful tool that is sufficient for precise data assessment. MCNP calculations were an indispensable part of our experiment technique and directly support analysis of the data. This paper reports the results of indium activation measurements and their role in radial neutron distribution (RND) measurements, analysis, and R_{as} determination. The SNEC is constructed from two other parameters, the SNEC with respect to the device axis R_s and the rotation of the emission center around the axis α_s within the PF-1000 device. This SNEC and R_{as} are newly introduced parameters.

Materials and methods

Plasma focus device and its surroundings

The PF-1000 is a Z-pinch-type device that produces hot, dense plasmas [3–5]. The plasma temperature is high enough for fusion reactions to take place. In addition, deuterium ions are accelerated by the magnetic field and collide with each other to produce intense streams of neutrons. In this way, the induced plasma emits large pulses of neutrons, up to 10^{12} per pulse [6]. These neutrons are emitted in a few characteristic directions and carry not only energy but also information concerning their production processes. In Fig. 1 one can see a view of the device from the side, while Fig. 2 shows the PF-1000 electrodes from the front of the device. The footprint of the plasma and blast wave on the front of the anode is irregular and asymmetric. Although this is not the result of a single

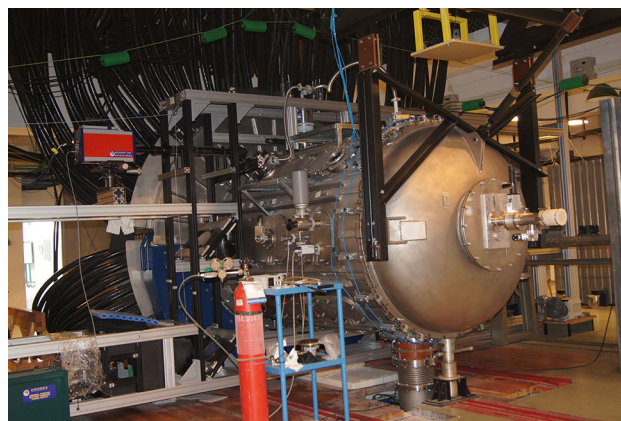


Fig. 1 Side view of the PF-1000 device

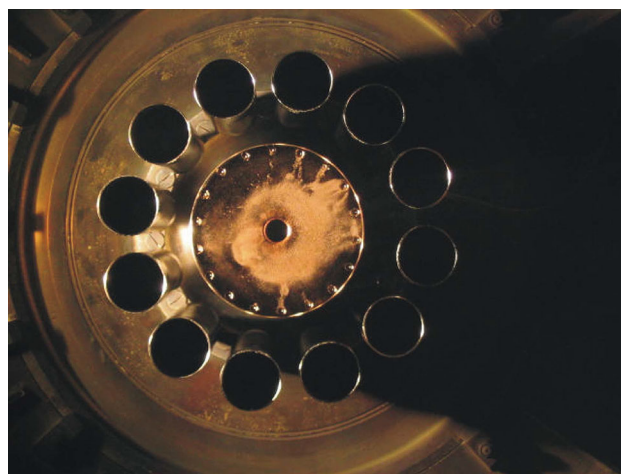


Fig. 2 PF-1000 electrode set: central anode surrounded by 12 stainless steel rods

shot, but rather of a series of fired discharges, it shows a lack of symmetry that we attempt to account for in our experiments.

IPPLM has several diagnostics that allow the monitoring of plasma discharge events both in real time and through offline analysis [7]. The activation technique is not a real-time diagnostic but is one of the most essential diagnostics at the facility [8, 9]. Indium activation belongs to this category as well.

The radiation that occurs during plasma discharge encounters on its way lot of barriers. The first shielding that any radiation released from the plasma column meets on its way is the vacuum vessel wall. The PF-1000 device chamber is fabricated of stainless steel 1H18N9T ($\rho = 7,9 \text{ g cm}^{-3}$). The wall thickness is 1 cm however there are many constructional details on its surface like longitudinal reinforcement ribs, diagnostic window frames, flanges,

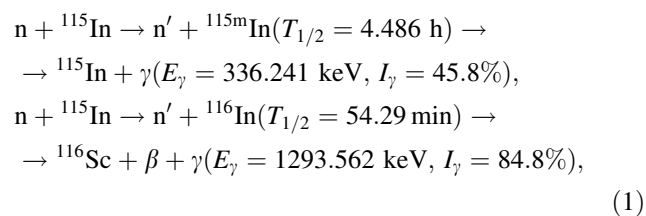
diagnostics, machinery components, etc., that together moderates penetrating radiation leaving the vacuum vessel. It is considered that PF-1000 walls are an effective shielding against electromagnetic ionizing radiation with energies below 70 keV, totally reducing the fluxes of charged particles and slightly influencing on other penetrating radiation.

The system of movable shielding mainly against neutrons was installed in the surroundings of PF-1000 device. The particular shielding has dimensions $200 \times 100 \times 20$ cm. It is made from paraffin covered by trapezoidal steel sheet, thickness 0.5 mm. The paraffin used was enriched with 18 % of borax. Regarding the standard procedure implemented on PF-1000 facilities moving the paraffin shielding during experimental campaign it is strongly prohibited.

The shielding system as well as other items present in the experimental hall influences on neutron spectrum by enriching it with a slowed down component. Thus the architecture of the experimental hall is significant when activation results in radiative capture e.g. $^{115}\text{In}(n,\gamma)^{116}\text{In}$ and insignificant influence when activation results in threshold reaction $^{115}\text{In}(n,n')^{115\text{m}}\text{In}$. During the experiment both indium isotopes were detected while in subsequent assessment only ^{115}In was considered.

Indium

Indium is used as an activation material in many large plasma experiments in order to monitor the total Y_n . $^{115\text{m}}\text{In}$ is also used for measuring the spatial distribution of neutron emission during plasma experiments performed on the PF-1000 device [2]. ^{116}In is formed simultaneously with $^{115\text{m}}\text{In}$ and has a negative effect on the accuracy of $^{115\text{m}}\text{In}$ detection. These two isotopes are the result of the following nuclear reactions:



where $T_{1/2}$ is the reaction half-life, E_γ is the energy of the most intense γ transition, and I_γ is the yield of the most intense γ transition.

The cross section of the threshold nuclear reaction $^{115}\text{In}(n,n')^{115\text{m}}\text{In}$ rapidly increases when the incident neutron energy exceeds ~ 320 keV. It reaches its maximal value (~ 0.36 b) at 2.70 MeV and after that slowly decreases. For the 2.45 MeV neutrons that are the subject of our interest, the

cross section is still relatively significant (~ 0.34 b) [10]. The cross section for neutrons with energies above 2.70 MeV does not have any meaning, owing to the limited energy of neutrons produced by PF-1000 (2.45 MeV in the system center of mass). Thus, the reaction is sensitive to D–D neutrons that leave the plasma with relatively high energy and insensitive to slowed-down neutrons.

The radioactive $^{115\text{m}}\text{In}$ is a source of γ -radiation. It has a relatively long half-life that allows us to measure a row of inactivated samples after their activation. But the threshold reaction occurs simultaneously with radiative capture reactions under the influence of neutrons emitted from plasma. The product of a radiative capture reaction (e.g., ^{116}In) is undesired and usually more radioactive than $^{115\text{m}}\text{In}$. It does not allow the immediate measurement of the $^{115\text{m}}\text{In}$ radioactivity. The 336.24 keV peak of the full energy absorption of γ -quanta emitted from $^{115\text{m}}\text{In}$ nuclei is surrounded by Compton background signal. Thus, only after cooling of the activated indium sample the correct estimation of $^{115\text{m}}\text{In}$ radioactivity is possible.

Gamma spectrometry

An inherent part of the activation technique is radiometry of activity induced by neutrons in a particular material. Contemporary accessible γ -spectrometers are equipped with high-efficiency, high-resolution HPGe detectors that allow very precise radioactivity determinations.

The measurements of radioindium were completed with a γ -spectrometry system consisting of a HPGe coaxial detector system with 30 % relative efficiency (FWHM = 1.80 keV for the 1332.49 keV γ -line) and an InSpector 2,000 multichannel analyzer. The data acquisition and analysis were performed with the Genie 2,000 Spectroscopy Software. The quantum, that was most important to the experiment, was the numerical characterization of the detector allowing very precise detector energy–efficiency calibration [11, 12].

After activation, the indium samples were collected manually, cooled down, and measured for 1000 s. The radioactivity was calculated for the moment of discharge. The computer systems were synchronized using the radio check clock. The minimum detectable activity (MDA) was in the range of 0.47 Bq per sample [13].

Optimization of sample geometry

The HPGe detector met the requirement of the activated sample measurement precision only when the registration efficiency for a particular γ -quantum was high enough. But when the number of activating neutrons is low, the only way to increase the sample activity is to increase the

sample mass in a specific way. During the sample optimization, both the sample's ability to be activated and the efficiency of particular γ -quanta (336.24 keV) registration in the HPGe spectrometry system were considered.

As a result of the optimization process, the indium RND monitor was proposed to be a heavy indium sample with a diameter of 63.7 mm and 4.3 mm in thickness. The efficiency calibration curve for this sample was generated as a result of a mathematical energy–efficiency calibration [14, 15].

MCNP calculation

The most effort-consuming part of the neutron analysis for calculating the fluence of neutrons in the samples lies in providing an accurate description of the geometric and material components of the device in the code. A geometric model of the PF-1000 facility has been prepared for neutron transport Monte Carlo calculations with the MCNP code. The structure of the steel chamber, steel and copper electrode set, and indium samples have been quite accurately modeled. The MCNP5 [16] code with the MCNP5DATA [17] cross-section library have been used for calculations of neutron distributions outside the chamber of the PF-1000 facility. Monte Carlo simulations were performed for 2.45 MeV neutrons. It was assumed that the neutrons were emitted from an isotropic point source placed at the center of the pinch plasma. The number of simulated histories for each neutron source was 2×10^8 . The estimated relative errors for the calculated results are usually below 1 %. The aim of the MCNP simulation was to assess the activation coefficient and the number of neutrons that induced 1 Bq of ^{115m}In in the sample material.

Radial neutron distribution measurement

Sample activation took place outside of the machine. This activation method gave information about the discrete distribution of neutrons during a single discharge. But the method was sensitive only when the Y_n exceeded a certain minimum value ($\sim 5 \times 10^9$ neutrons per pulse). After activation, indium samples were measured using γ -spectrometry.

For the RND measurements, a set of eight indium samples was used. They were distributed regularly on the cylindrical part of the vacuum vessel outer wall. All samples were deployed 75.1 mm from the pinch on the plane that crossed the device's axis and the plasma pinch perpendicularly. This plane was parallel to the anode surface and removed from its surface by 6 cm. Each sample position was rotated by 45° relative to the others. In Fig. 3 is presented the sample configuration over the external

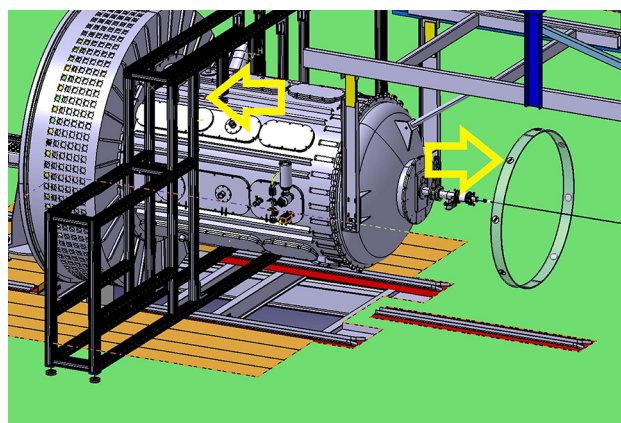


Fig. 3 Design drawing of PF-1000 with the isolated schematic view of the ring (virtually cut out from the device) with a set of eight samples (right arrow). The samples were placed on the outer vacuum vessel wall, on the plane perpendicular to the device axis. Only one indium sample is visible (left arrow) (drawn in CATIA). (Color figure online)

circuit of the PF-1000 device. While only one indium sample is visible on the device outer wall (left yellow arrow), the virtual ring with eight samples (right yellow arrow) represents their composition. In total, 104 measurements of the RND (13 series of 8 points) were completed.

The results of the RND measurements are shown in Fig. 4. The experimental data points were fitted with a polynomial function. They have the shape of closed curves in the polar coordinates. The area of the region bounded by each curve was calculated as a result of integration using a definite Riemann integral. Integration was carried out numerically. After that, the relationship between areas and corresponding values of Y_n was examined. The numerical value of the integral for the j th discharge was the total number of neutrons emitted per 2π angle that lay on the plane perpendicular to the PF-1000 device and crossed the plasma pinch through its hypothetical center. The n_{Av} emission was calculated by dividing the value of integral by the limit of integration. Thus n_{Av} is the neutron emission into the solid angle of the sample. The n_{Av} was also calculated as the arithmetic mean of neutrons that crossed all indium samples ($i = 1-8$) during the j th discharge.

Based on the integrated and average values of neutron emission, the R_{as} coefficient was calculated using the following formula:

$$R_{an}^j = \sum_{i=1}^8 \frac{|n_{Av}^j - n_i^j|}{n_{Av}^j} \quad (2)$$

where R_{an}^j is the R_{as} of transverse neutrons distribution recorded during the j th discharge, n_{Av}^j is the average value of the neutron flux (neutrons per pulse) that crossed all

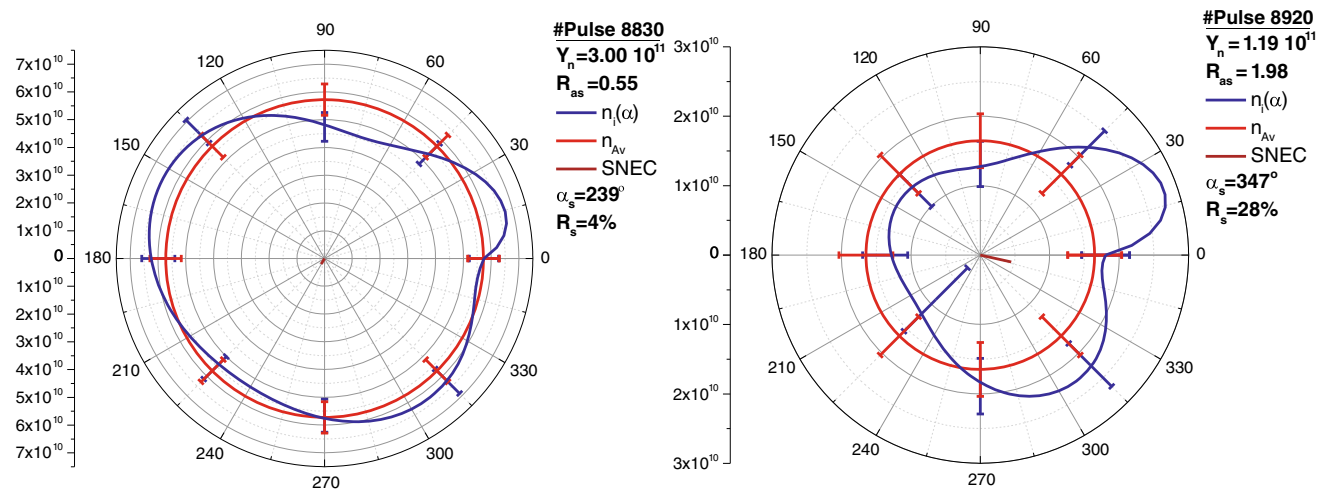


Fig. 4 RND for two cases with extreme values of R_{as} . The used parameters are Y_n the measured total neutron yield, R_{as} the radial asymmetry, $n_i(\alpha)$ the fluence in the sample i th lying on the angle α (blue line), n_{Av} the average neutron fluence in the j th discharge (red

line), SNEC the shift of a neutron emission center (brown line), R_s a segment representing the shift of the emission center across the main device axis, and α_s rotation of the emission center around the axis of the PF-1000 device (drawn in OryginPro 8.5). (Color figure online)

activation samples during the j th discharge, and $n_i^j(\alpha)$ is the neutron flux that crossed the i th indium sample during the j th discharge.

R_{as} is change from $<0, \infty$). When R_{as}^j is equal to or very close to 0, the neutron distribution shape is equal or very close to circular. Conversely, when the neutron distribution is more deformed, the R_{as} is more pronounced. In these calculations the R_{as} did not exceed two. A case where neutron emission in one particular direction is significantly higher than in others is physically unreasonable. The R_{as} is the measured quantity that allows characterization of the transverse neutron emission.

Next, the SNEC was calculated for each observed RND. The interpretation of the SNEC is that it is the motion of point that represents the center of each particular RND in relation to the device axis. In polar coordinates it is expressed by the following parameters: the radial shift (R_s) which illustrates the movement of the hypothetical center of neutron emission across the device axis, and the angle of rotation (α_s) that quantifies the revolution of the above point around the device axis. The R_s is presented as part of n_{Av} and expressed as a percentage. The SNEC and its parametric representation are useful for characterization of plasma experiments performed at the PF-1000 facility.

Results

The MCNP calculations give two important pieces of information regarding plasma formation in the PF-1000 device. First, the activation coefficient is equal to 6.39×10^{-30} , representing the number of indium nuclei

activated by a single D–D neutron. Second, the number of source neutrons that produce 1 Bq of ^{115m}In in the irradiated sample is equal to 7.3×10^9 neutrons. This second parameter allows us to express the result of indium activation as the number of neutrons that leave the plasma in different α -directions (occupied by an indium sample).

The calculated values of n_{Av} were compared with Y_n as measured by a calibrated silver activation counter. The results were linearly dependent on each other.

The polar graphs that show the results of measurements and calculation are presented in Fig. 4. There are two curves on each figure, and both of them present different cases of the RND. The red line shows the idealistic representation of n_{Av} and the blue line is the polynomial fit of $n_i^j(\alpha)$. In the key of each figure the relevant parameters are listed for each measurement. The precise values with their uncertainty for the RND experiment campaign are summarized in Table 1.

The data for R_{as} , the main subject of this work, are presented in Fig. 5. They show a trend in that higher Y_n gives a lower R_{as} . But some results did not follow this trend (in particular, the two outliers at lower left in Fig. 5). It is not fundamentally possible to explain the outliers at present.

In Fig. 6 is shown the SNEC represented by R_s as a function of Y_n . The R_s is sensitive as the parameter describing R_{as} only when it manifested itself by asymmetric deformation. When the neutron emission shape is flattened on both sides, the R_s is relatively low and the R_{as} is high. Conversely, when the RND graph is circular but shifted with respect to the graph that represents the averaged neutron's emission, the R_s is high and the R_{as} is low.

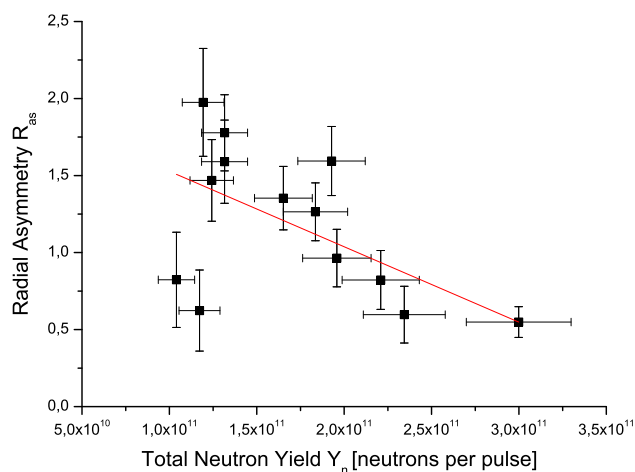
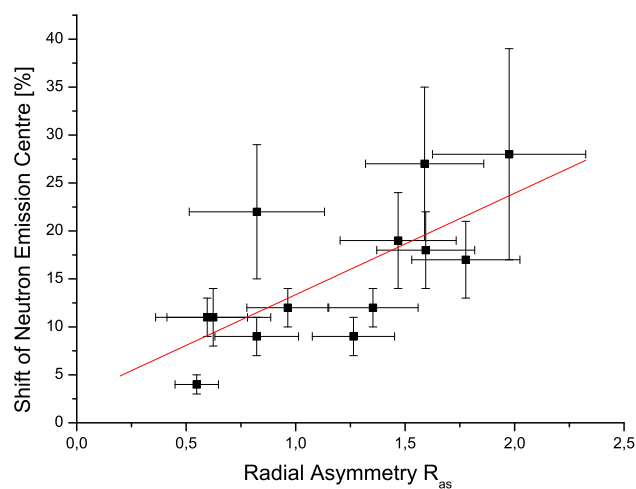
Table 1 Summary of the measured experimental parameters

Pulse no	$Y_n \times 10^{11}$ [neutrons $\times$ pulse $^{-1}$]*	Integral $\times 10^{12}$ [neutrons per 2π angle]	$n_{AV} \times 10^{10}$ [neutrons $\times$ solid angle]	R_{as}^{**}	SNEC	
					R_s [%]	α_s [degree]
#8,824	1.04	5.93 ± 1.29	1.64 ± 0.35	0.82 ± 0.31	22 ± 7	135 ± 43
#8,915	1.17	7.15 ± 1.33	1.98 ± 0.37	0.62 ± 0.26	11 ± 3	176 ± 47
#8,920	1.19	5.85 ± 1.42	1.62 ± 0.39	1.98 ± 0.35	28 ± 11	347 ± 136
#8,939	1.24	7.32 ± 1.36	2.03 ± 0.38	1.47 ± 0.27	19 ± 5	164 ± 45
#8,821	1.32	8.57 ± 1.60	2.38 ± 0.44	1.59 ± 0.27	27 ± 8	79 ± 25
#8,935	1.32	7.85 ± 1.36	2.18 ± 0.37	1.78 ± 0.25	17 ± 4	216 ± 56
#8,825	1.65	10.43 ± 1.51	2.89 ± 0.42	1.35 ± 0.21	12 ± 2	150 ± 32
#8,953	1.84	12.14 ± 1.60	3.37 ± 0.50	1.26 ± 0.19	9 ± 2	233 ± 45
#8,940	1.93	9.49 ± 1.50	2.63 ± 0.41	1.59 ± 0.22	18 ± 4	211 ± 49
#8,923	1.96	12.4 ± 1.64	3.44 ± 0.45	0.96 ± 0.19	12 ± 2	208 ± 39
#8,951	2.21	11.94 ± 1.60	3.31 ± 0.44	0.82 ± 0.19	9 ± 2	59 ± 11
#8,949	2.34	13.23 ± 1.72	3.67 ± 0.47	0.60 ± 0.18	11 ± 2	276 ± 51
#8,830	3.00	20.58 ± 2.04	5.71 ± 0.56	0.55 ± 0.10	4 ± 1	239 ± 34

Y_n is the total neutron yield, the Integral is the neutron emission per 2π angle, n_{AV} is the neutron emission into the solid angle of the sample, R_{as} is the radial asymmetry, SNEC(R_s , α_s) is the shift and rotation of the neutron emission center

* The uncertainty of Y_n determination is $\sim 10\%$ and comes from the uncertainty of the calibration factor which is constant and equal to 10% and the number of counts recorded by silver counter (from 1×10^2 up to 2×10^5) after particular discharge

** Unitless

**Fig. 5** Radial asymmetry as a function of Y_n (drawn in OryginPro 8.5)**Fig. 6** SNEC represented by R_s as the function of R_{as} . R_s is presented as the part of n_{AV} and expressed in % (drawn in OryginPro 8.5)

Thus, the universality of R_s parameter is limited in comparison with R_{as} .

Conclusions

The activation technique was successfully adapted to RND measurements for the PF-1000 device. The nuclear properties of indium, such as its cross section and ^{115}In half-

life, allowed determination of the discrete character of RND.

The parameter R_{as} allowed qualitative and quantitative assessment of the RND of neutrons that were emitted during plasma experiments conducted on the PF-1000 device. When the R_{as} value was close to 0.5, the neutron emissions were near-uniform in all directions and the radial distribution of the emission was close to circular (see pulse 8,830 in Fig. 4). However, when the R_{as} approached two,

there was significant irregularity in the neutron emission direction, and the radial distribution of the emission was strongly deformed and asymmetric (see pulse 8,920 in Fig. 4). A trend was observed for more results that show that a higher total Y_n corresponded to a smaller R_{as} .

The RND measurements have shown that source of neutrons has different shape in each separate discharge. The expected circular shape was only measured in a limited number of plasma discharges. This means that the neutron source had an irregular shape and could fluctuate across the device axis. This fluctuation was measured and recorded as the R_{as} and SNEC, represented by R_s and α_s . Irregularity in the RND shape means that there was some directional instability in the neutron emission, skewed towards certain directions. This phenomenon may be explained by the presence of instabilities. When pinched plasmas have high levels of impurities, they can develop instabilities ($m = 0$) that results in radiative collapse and formation of so-called hot-spots [1]. But the origin of this observed phenomenon is unconfirmed.

The areas of the shapes representing the RND were linearly proportional to Y_n even though the shapes of RND were irregular. There was no simple correlation between R_s and R_{as} , though one could see a trend where the higher R_{as} resulted in a higher R_s (see Fig. 6). α_s did not correlated with any other parameter. The value of R_s may have been somehow limited by the diameter of the electrodes, but this was not studied in this campaign.

In the case of large R_{as} , monitoring of the total Y_n and forward anisotropy by a one-point method (such as the silver activation counter) may lead to erroneous results.

The precision of the obtained results was limited by the uncertainty of the radiometric measurement immediately after low Y_n pulses. The accuracy of our assessments could be increased only by changing the HPGe detector for one with higher relative efficiency.

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References

- Bernard A, Bruzzone H, Choi P, Chuaqui H, Gribkov V, Herrera J, Hirano K, Krejci A, Lee S, Luo C, Mezzetti F, Sadowski M, Schmidt H, Ware K, Wong CS, Zoita V (1998) Scientific status of plasma focus researches. *J Moscow Phys Soc* 8:93–170
- Jednorog S, Szydowski A, Scholz M, Paduch M, Bienkowska B (2012) Preliminary determination of angular distribution of neutrons emitted from PF-1000 facility by indium activation. *Nukleonika* 57(4):563–568
- Schmidt H, Kasperczuk A, Paduch M, Pisarczyk T, Scholz M, Tomaszewski K, Szydowski A (2002) Review of recent experiments with the megajoule PF-1000 plasma focus device. *Phys Scr* 66(2):168–197
- Bienkowska B, Karpinski L, Paduch M, Scholz M (2006) Measurements of neutron yield from PF-1000 device by activation method. *Czech J Phys* 56:B377–B382
- Szydowski A, Scholz M, Karpinski L, Sadowski M, Tomaszewski K, Paduch M (2001) Neutron and fast ion emission from PF-1000 facility equipped with new large electrodes. *Nukleonika* 46(Suppl 1):S61–S64
- Scholz M, Bienkowska B, Chernyshova M, Gribkov V, Jednorog S, Kalinowska Z, Karpinski L, Paduch M, Prokopowicz R, Szydowski A (2007) Fast-neutron source based on Plasma-Focus device. *Proc of the Int Conf on Researches and Applications of Plasmas PLASMA 2007*. Greifswald, Germany, pp 345–348
- Jednorog S, Szydowski A, Bienkowska B, Prokopowicz R (2014) The application of selected radionuclides for monitoring of the D–D reactions produced by dense plasma-focus device. *J Radioanal Nucl Chem*. doi:10.1007/s10967-014-3131-0
- Bienkowska B, Karpinski L, Paduch M, Scholz M (2006) Measurements of neutron yield from PF-1000 device by activation method. *Czech J Phys* 56:B377–B382
- Bienkowska B, Scholz M, Wincel K, Zareba B (2007) Use of activation technique and MCNP calculations for measurement of fast neutron spatial distribution at the MJ Plasma Focus device. *Proc of the Int Conf on Researches and Applications of Plasmas PLASMA 2007*. Greifswald, Germany, pp 349–352
- International Atomic Energy Agency (2012) International Reactor Dosimetry File 2012, Technical Reports Series 452 VIENNA
- Bronson FL, Venkataraman R (2000) Validation of the accuracy of the LabSOCS mathematical efficiency calibration for typical laboratory samples, 46th Annual Conference on Bioassay, Analytical, and Environmental Radiochemistry, 12–17 Nov 2000, Seattle, Washington DC
- Bronson FL, Venkataraman R, Young B (1998) Massometric efficiency calibrations of HPGe detectors for Laboratory applications 44th Annual Conference on Bioassay, Analytical, and Environmental Radiochemistry, 15–19 Nov 1998, Albuquerque NM
- Glimore G (2008) Practical gamma-ray spectrometry, 2nd edn. Wiley, Chichester
- Jednorog S, Scholz M, Popovichev S, Murari A, JET EFDA contributors (2009) Numerical Optimization of Activation Samples for the Application of the Activation Technique to Measure Neutrons in Large Fusion Devices like JET and ITER, EFDA–JET–CP (09) 06/03: 1–5
- Ciupek K, Jednorog S, Fujak M, Szewczak K (2014) Evaluation of efficiency for in situ gamma spectrometer based upon cerium-doped lanthanum bromide detector dedicated for environmental radiation monitoring. *J Radioanal Nucl Chem* 299:1345–1350
- X-5 Monte Carlo Team (2003) MCNP—a general Monte Carlo N-particle transport code
- X-5 Monte Carlo Team (2003) MCNP5DATA: Standard Neutron Photoatomic, Photonuclear, and Electron Data Libraries for MCNP5 (CCC-710)